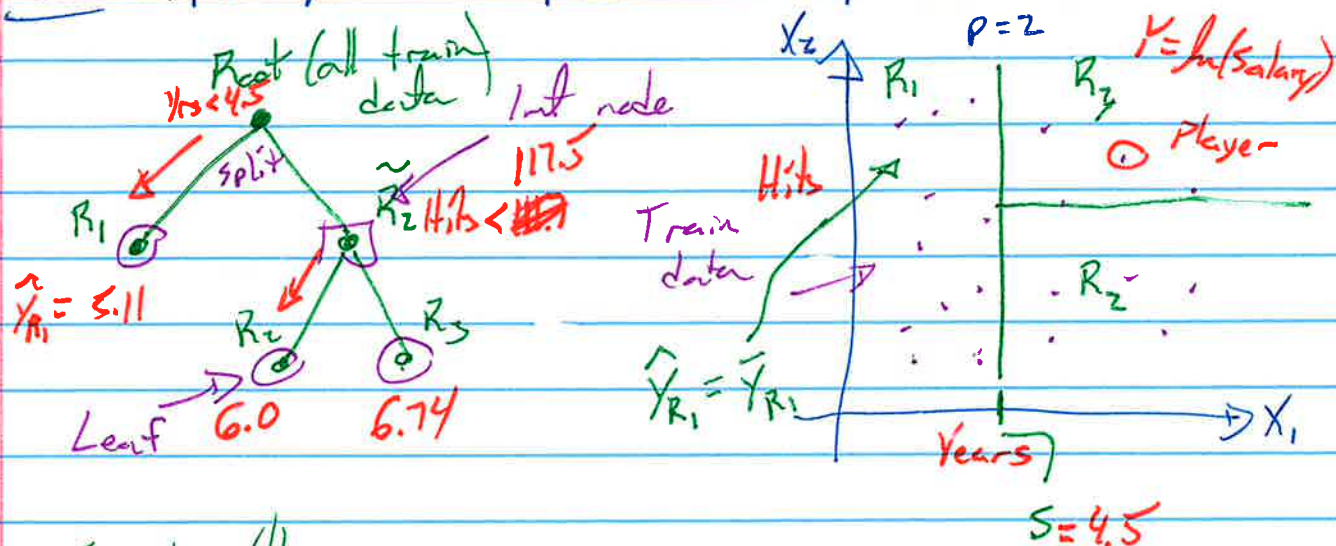


CH8 Tree based learning Decision tree basics ✓
 Ensembles of trees (many) (bagging random forest boosting)

Regression case $Y = f(X) + \epsilon$, $X = (X_1, X_2, \dots, X_p)^T$

Idea: Split predictor space to separate Y



DT algorithm

① Consider all splits s on X_j into 2 regions

$$R_1(j, s) = \{X \mid X_j \leq s\}$$

$$R_2(j, s) = \{X \mid X_j > s\}$$

② Choose s & X_j which minimize

$$RSS = \sum_{X \in R_1} (Y_i - \hat{Y}_{R_1})^2 + \sum_{X \in R_2} (Y_i - \hat{Y}_{R_2})^2$$

③ Iterate ① & ② on each region until some stop condition is met. 2 options.
 (i) Heuristic (leaf population below threshold)
 (ii) Tree depth via CV

Tree size selection via CV:

Issue: Grow tree completely (one point per leaf)
needs 2^n branches. Too much compute.

Use LASSO ish idea to regularize tree depth.

(Cost complexity pruning)

$$CCP_\alpha(T) = RSS + \alpha |T|$$

↑
full grown tree
leaf count

$\alpha \geq 0$ CV parameter to tune (ish)

DT Pros/cons

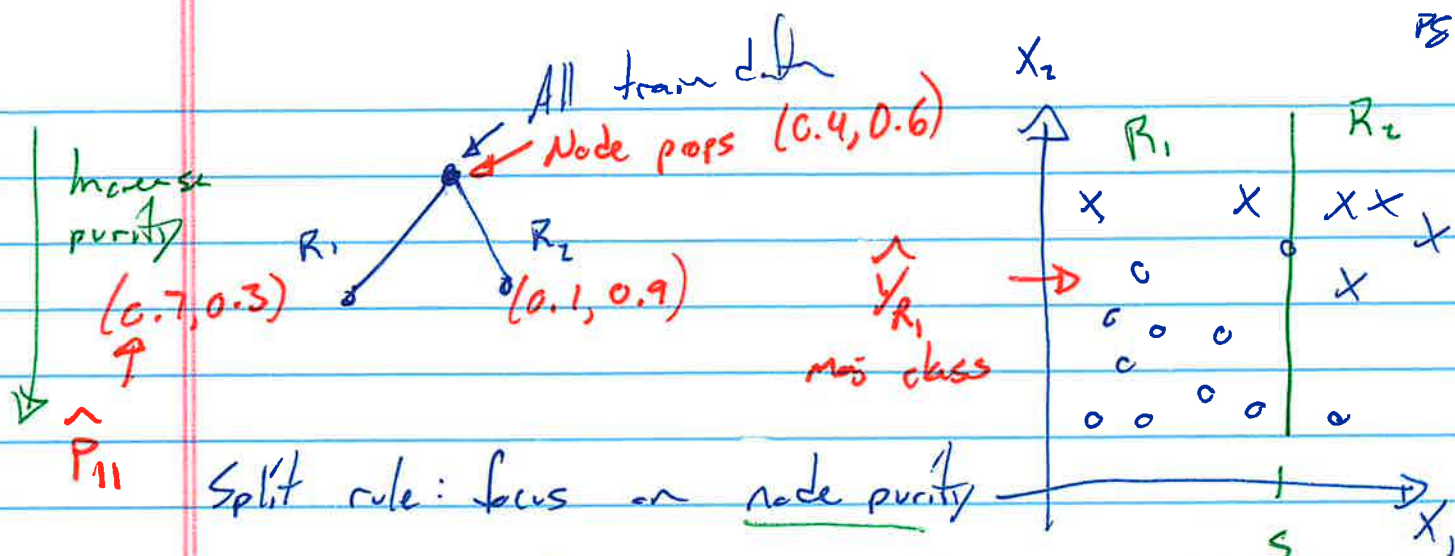
Pros: Simple & readable, data-driven (no model assumed non-parametric)
Low bias

Cons: High variance, low accuracy (LR much better)
↳ Fix via ensemble methods.

Classification trees $Y \in \{0, 1\}$, $X = (X_1, X_2, \dots, X_p)^T$

Almost same as linear regression.

2 differences: ① Prediction $\bar{y}$ (sad face)
② Split rule RSS (sad face)



Split rule: focus on node purity

3 ways: ① Classification error = $1 - \max_k \hat{p}_{mk}$
 Isn't strictly end on purity → Want small classes → Node m

② Gini index = $G_m = \sum_k \hat{p}_{mk} (1 - \hat{p}_{mk})$
 Total var explained →

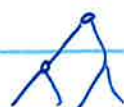
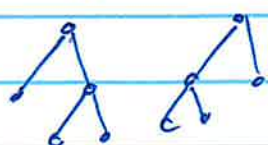
③ Entropy = $-\sum_k \hat{p}_{mk} \ln(\hat{p}_{mk})$
 Infer theory → measure disorder

For all splits (i, s) , find best increase in purity for split choice.

True Any model ensembles Solve issue of 1 tree high variance (low accuracy)

Average many trees (short simple) T_N ($N=1000$)

Hard to interpret →



Variable (avg) imp. (RRS dec purity int)

$$\hat{y} = \begin{cases} \text{avg all } T_i \hat{y} & (\text{reg}) \text{ SD}(\hat{y}) \\ \text{maj class } T_i \hat{y} & (\text{classif}) \text{ Class props} \end{cases}$$

heavy background

PS 9

Why does ensemble reduce variance?

(145) Start: Mean estimate

$$Z \sim N(\mu, \sigma^2)$$



Rough Idea.

many trees
Samples: $Z_1, Z_2, \dots, Z_n$

$$\mu \hat{Z} = \frac{Z_1 + \dots + Z_n}{n}, \quad SE(\hat{Z}) = \frac{\sigma^2}{n}$$

ensemble

new var decreases.

2 ensemble approaches: ① Bootstrap → Bossing / RF
② Recursive (on residuals) → Boosting.

Bootstrap ensembles

Generate many trees

ensemble avg $(1 - 1/n)^n$

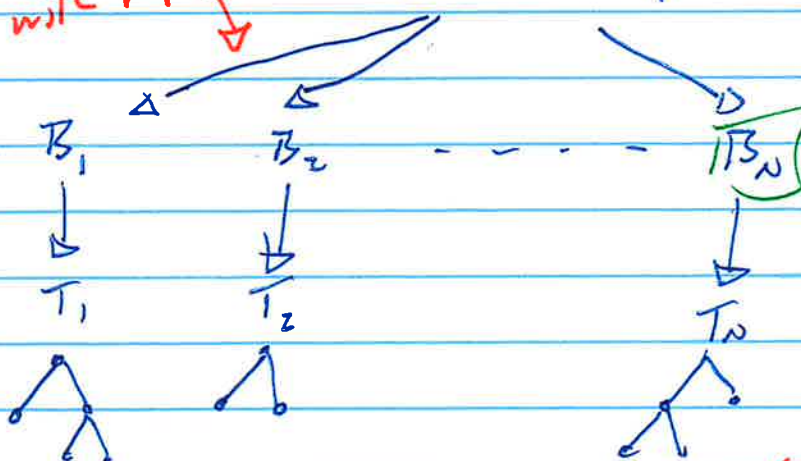
Bossing (bootstrap aggregating) idem.

Bootstrap, sample with replacement

Train sample

Past HW showed $\sim 1/3$ train data not in B_i .

Keep as test data per tree.



OOB samples

out of bag
(Free CV in process)
tune on tree depth d .

Bagging issue: Trees often highly correlated especially for large train data.

B_i keeps $\sim 2/3$ train data.

Result is little decrease in model variance.

⊛ Random forest fixes this!

↳ Same as bagging, but for each split only consider some predictors $X_1, X_2, \dots, X_p$

$m = \sqrt{p}$ random predictors

Result: Much higher accuracy than bagging. (see text)

Recursive or residuals

Boosting

Learns what

T_1 misses

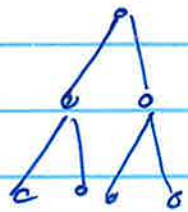
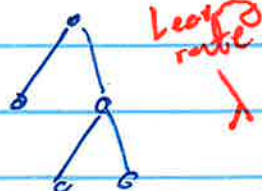
" "

T_2 misses

$T_1 (y \approx \hat{y})$

$T_2 (r_1 \approx \hat{r}_i)$

T_3



.....

$$r_{1i} = y_i - \hat{y}_i$$

$$r_{2i} = r_{1i} - \hat{r}_{1i}$$

↑ residual

Also messy-ish.

↳ small force many trees chosen via CV